

SPECIAL DISCRETE RANDOM VARIABLES

CHAPTER 03

Bernoulli Random Variable

CONT.

- A Bernoulli random variable is the simplest kind of random variable.
- A Bernoulli trial is an experiment in which the outcome is either a success or a failure.(It can take on two values, 1 and 0.)
- It takes on a 1 if an experiment with probability p resulted in success and a 0 otherwise.
- Ex: coin flip, a random binary digit, whether a disk drive crashed, and whether someone likes a Netflix movie.

Cont.

- If X is a Bernoulli random variable, denoted $X \sim \text{Ber}(p)$:

Probability mass function:	$P(X = 1) = p$
	$P(X = 0) = (1 - p)$
Expectation:	$E[X] = p$
Variance:	$\text{Var}(X) = p(1 - p)$

- Bernoulli random variables and indicator variables are two aspects of the same concept.
- A random variable I is called an indicator variable for an event A if $I = 1$ when A occurs and $I = 0$ if A does not occur.
- $P(I = 1) = P(A)$ and $E[I] = P(A)$. Indicator random variables are Bernoulli random variables, with $p = P(A)$.

Example 1

- When $p = 1/2$ (e.g., for tossing a coin), we have
$$E[X] = p = 1/2, \text{Var}(X) = p(1 - p) = 1/4.$$
- When rolling a die, with outcome k , ($1 \leq k \leq 6$),
let $X(k) = 1$ if the roll resulted in a six, and $X(k) = 0$ if the roll did not result in a six.
Then $E[X] = p = 1/6, \text{Var}(X) = p(1 - p) = 5/36$
- When $p = 0.01$, then $E[X] = 0.01, \text{Var}(X) = 0.0099 \approx 0.01$.

Binomial Random Variable

Cont.

- A binomial random variable is random variable that represents the number of successes in n successive independent trials of a Bernoulli experiment.
- Ex: the number of heads in n coin flips, the number of disk drives that crashed in a cluster of 1000 computers, and the number of advertisements that are clicked when 40,000 are served.

Definition

- Suppose a random experiment has the following characteristics.
- There are n identical and independent trials of a common procedure.
- *There are exactly two possible outcomes for each trial, one termed “success” and the other “failure.”*
- The probability of success on any one trial is the same number p .
- Then the discrete random variable X that counts the number of successes in the n trials is the **binomial random variable with parameters n and p** . We also say that X **has a binomial distribution with parameters n and p** .

Examples

- A random sample of 125 students is selected from a large college in which the proportion of students who are females is 57%. Suppose X denotes the number of female students in the sample. In this situation there are $n = 125$ identical and independent trials of a common procedure, selecting a student at random; there are exactly two possible outcomes for each trial, “success” (what we are counting, that the student be female) and “failure;” and finally the probability of success on any one trial is the same number $p = 0.57$. X is a binomial random variable with parameters $n = 125$ and $p = 0.57$.
- A multiple-choice test has 15 questions, each of which has five choices. An unprepared student taking the test answers each of the questions completely randomly by choosing an arbitrary answer from the five provided. Suppose X denotes the number of answers that the student gets right. X is a binomial random variable with parameters $n = 15$ and $p = 1/5 = 0.20$.

Examples

- In a survey of 1,000 registered voters each voter is asked if he intends to vote for a candidate Titania Queen in the upcoming election. Suppose X denotes the number of voters in the survey who intend to vote for Titania Queen. X is a binomial random variable with $n = 1000$ and p equal to the true proportion of voters (surveyed or not) who intend to vote for Titania Queen.
- An experimental medication was given to 30 patients with a certain medical condition. Suppose X denotes the number of patients who develop severe side effects. X is a binomial random variable with $n = 30$ and p equal to the true probability that a patient with the underlying condition will experience severe side effects if given that medication.

Cont.

- If X is a Binomial random variable, we denote this

$$X \sim \text{Bin}(n, p),$$

where p is the probability of success in a given trial.

- A binomial random variable has the following properties:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad \text{if } k \in N, 0 \leq k \leq n \text{ (0 otherwise)}$$

$$E[X] = np$$

$$\text{Var}(X) = np(1 - p)$$

Proof

- If in a sequence of n independent Bernoulli trials
we let X_k = the outcome of the k^{th} Bernoulli trial , ($X_k = 0$ or 1) ,
then $X \equiv X_1 + X_2 + \dots + X_n$, is the Binomial random variable that counts the successes ” .

$$X \equiv X_1 + X_2 + \dots + X_n$$

- We know that $E[X_k] = p$,
so $E[X] = E[X_1] + E[X_2] + \dots + E[X_n] = np$.
- We already know that $Var(X_k) = E[X_k^2] - (E[X_k])^2 = p - p^2 = p(1 - p)$,
so, since the X_k are independent , we have
$$Var(X) = Var(X_1) + Var(X_2) + \dots + Var(X_n) = np(1 - p) .$$
- NOTE : If p is small then $Var(X) \cong np$.

Example 2

- Let X = number of heads after a coin is flipped three times. $X \sim \text{Bin}(3, 0.5)$. What is the probability of each of the different values of X ?

$$P(X = 0) = \binom{3}{0} p^0 (1 - p)^3 = \frac{1}{8}$$

$$P(X = 1) = \binom{3}{1} p^1 (1 - p)^2 = \frac{3}{8}$$

$$P(X = 2) = \binom{3}{2} p^2 (1 - p)^1 = \frac{3}{8}$$

$$P(X = 3) = \binom{3}{3} p^3 (1 - p)^0 = \frac{1}{8}$$

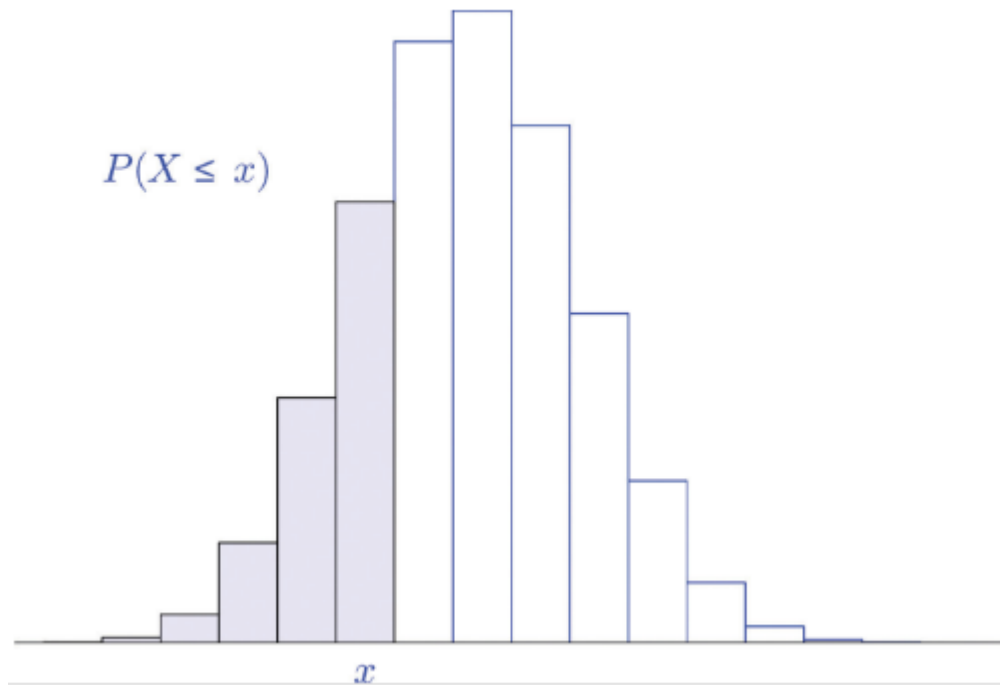
Example 3

Seventeen percent of victims of financial fraud know the perpetrator of the fraud personally.

- I. Use the formula to construct the probability distribution for the number X of people in a random sample of five victims of financial fraud who knew the perpetrator personally.
- II. A investigator examines five cases of financial fraud every day. Find the most frequent number of cases each day in which the victim knew the perpetrator.
- III. A investigator examines five cases of financial fraud every day. Find the average number of cases per day in which the victim knew the perpetrator.
- IV. Find the mean and standard deviation of the random variable X .

The Cumulative Probability Distribution of a Binomial Random Variable

Cumulative Probabilities



If X is a discrete random variable, then

$$P(X \geq x) = 1 - P(X \leq x - 1)$$

and

$$P(x) = P(X \leq x) - P(X \leq x - 1)$$

Example 4

A student takes a ten-question true/false exam.

- I. Find the probability that the student gets exactly six of the questions right simply by guessing the answer on every question.
- II. Find the probability that the student will obtain a passing grade of 60% or greater simply by guessing.

Poisson Random Variable

Cont.

- A Poisson random variable gives the probability of a given number of events in a fixed interval of time (or space).
- It makes the Poisson assumption that events occur with a known constant mean rate and independently of the time since the last event.

Notation: $X \sim \text{Poi}(\lambda)$

Description: Number of events in a fixed time frame if (a) the events occur with a constant mean rate and (b) they occur independently of time since last event

Parameters: $\lambda \in \{0, 1, \dots\}$, the constant average rate.

Support: $x \in \{0, 1, \dots\}$

PMF equation: $P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$

Expectation: $E[X] = \lambda$

Variance: $\text{Var}(X) = \lambda$

PMF graph: _____

Definition

- Suppose an experiment is to be performed with the following characteristics:
- The random variable X can be thought of as a Binomial random variable with parameters n and p .
- n is sufficiently large; p is sufficiently small.
- Then $X \sim \text{Poisson}(\lambda)$ where $\lambda = np$.

Theorem

Suppose there is some event which occurs according to a Poisson distribution. That is, its occurrence can be modeled by $X \sim \text{Poisson}(\lambda)$ which means the event occurs at a mean rate of λ occurrences per unit. Then the random variable Y which models the number of events occurring in the interval of length t is given by $Y \sim \text{Poisson}(\lambda t)$.

Example 1

Show that the probabilities assigned by the Poisson probability distribution satisfy the requirements that $0 \leq p(y) \leq 1$ for all y and $\sum_{y=0}^{\infty} p(y) = 1$.

Because $\lambda > 0$, it is obvious that $p(y) > 0$ for $y = 0, 1, 2, \dots$, and that $p(y) = 0$ otherwise. Further,

$$\sum_{y=0}^{\infty} p(y) = \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} e^{-\lambda} = e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = e^{-\lambda} e^{\lambda} = 1$$

because the infinite sum $\sum_{y=0}^{\infty} \lambda^y / y!$ is a series expansion of e^{λ} .

Example 2

A manufacturer of light bulbs knows that 2% of its bulbs are defective. The manufacturer sells the bulbs in packs of 100. Assuming that the bulbs are independent of each other, find the probability that the pack contains at most three defective bulbs.

Example 3

Customers arrive at a restaurant at a rate of 10 customers per hour. Assuming that the number of arrivals follows a Poisson distribution, find the probability that more than 35 customers arrive within the span of three hours.

Homework

Find the moment-generating function $m(t)$ for a Poisson distributed random variable with mean λ .